

Harshana Lakshara Fernando

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Summary

Full-Stack Software Engineer experienced in end-to-end application delivery, legacy integrations, and production support. Strong in **Java**, **Spring Boot**, **Kafka**, **React**, and DevOps workflows. Known for rapid learning, taking ownership of distributed systems, and choosing solutions that balance reliability, scalability, and real product needs.

Skills

Core	Java Spring Boot Microservices Distributed Systems System Design REST APIs
Backend	Spring Cloud Spring Security Resilience4j Node.js Express.js Gin
Messaging & APIs	Kafka RabbitMQ gRPC GraphQL WebSockets Protobuf JWT OAuth 2.0
Data	PostgreSQL MySQL Oracle DB Redis Couchbase
Cloud & DevOps	AWS Docker Kubernetes CI/CD (GitHub Actions, Jenkins)
Testing	JMeter Gatling Pytest Unit Testing
Tools	Git Gradle Postman Jira Confluence

Experience

R&D Software Engineer — KōreroCare Platform

Mar 2026 – Present

Empathic Computing Laboratory, University of Auckland

Remote

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an **HL7/FHIR**-aligned data schema mapping participants, devices, and wearable measurements.

Software Engineer Intern — ISA Aviation Technology

Dec 2024 – May 2025

Information Systems Associates (Pvt) Ltd

Colombo, Sri Lanka

- Owned design and delivery of a distributed **Spring Boot/Kafka**-backed PNR ID generation service, **live in production**, achieving **25+ PNRs/sec** at **<100ms latency**.
- Designed a **pool-based PNR allocation model** to replace legacy Oracle-sequence generation, supporting short domain-specific PNR formats while avoiding predictable sequential identifiers.
- Implemented **concurrency-safe** PNR allocation using a **Kafka-backed pool**, ensuring the same PNR was not assigned to multiple service instances, while using Java **ReentrantLock** for application-level critical sections.
- Led the rollout from legacy PNR generation to AeroOrder using database-driven **feature toggles** and a **strangler-pattern** migration approach for phased production cutover and fallback.
- Resolved production **Kafka consumer lag** and offset-reset issues by adding graceful shutdown hooks, retry handling, validation, and a Producer-Consumer refactor.
- Integrated **gRPC** clients and Kafka consumers into the legacy **Java 8** AeroMart monolith for backward-compatible communication with AeroOrder microservices.
- Built async load-testing and monitoring support using **JMeter**, **Prometheus**, and **Grafana** to validate PNR/order flows, queue time, consumer behavior, and service latency.
- Added Jenkins environment support, maintained **>90% unit test coverage**, and coordinated delivery with QA and distributed engineering teams.

Freelance Software Engineer — Selected Client Projects

May 2025 – May 2026

Client Platforms

Remote / Sri Lanka

- Delivered Agrinova, a full-stack multi-branch inventory platform (Vercel, Supabase/PostgreSQL), implementing transaction-safe stock logic across 3 core workflows (issuance, returns, purchases).
- Reduced repeat dashboard data-fetch latency from ~384ms to <1ms for cached views using React Query caching and pagination.

- Owned end-to-end client-facing fixes, resolving production authentication, version compatibility, and workflow-level validation bugs.

Selected Projects

Employee Management System — Java, Spring Boot, Spring Cloud, Kubernetes

- Built cloud-native **Spring Boot** microservices using Eureka, Config Server, API Gateway, RabbitMQ-based configuration refresh, and Resilience4j fault tolerance.
- Added CI/CD, monitoring, Docker deployment, and Kubernetes-based architecture to improve scalability.

GoTogether Travel Platform — Next.js, Go, Spring Boot, Keycloak, PostgreSQL

- Built a full-stack travel platform with Next.js interfaces, Spring Boot social services, and **Go/gRPC** itinerary APIs.
- Implemented **OAuth2/OIDC** authentication with Keycloak and deployed services using Docker, NGINX, Certbot, and Oracle Cloud.

Event-Driven Food Delivery Platform — Java, Spring Boot, React, Kafka, Redis

- Built a full-stack food delivery platform with a React frontend, Spring Boot services, JWT authentication, and Redis caching.
- Implemented real-time order synchronization and notifications using **Kafka**, WebSockets, and service-to-service APIs.

Serverless Auction Platform — Node.js, AWS Lambda, DynamoDB, EventBridge

- Designed event-driven auction, authentication, and notification services using AWS Lambda, API Gateway, EventBridge, SQS/SNS, and DynamoDB GSIs.
- Secured APIs using a custom JWT Lambda Authorizer and improved reliability through validation and centralized error handling.

ZK-Rollup L2 Benchmarking Framework — Rust, Solidity, Python, Docker

- Built an end-to-end benchmarking pipeline with a Rust submitter, Solidity L1 contracts, and Python metrics and data-analysis tools.
- Increased submitter test coverage from **41% to 88%** and measured approximately **70% lower** data-availability cost using blob-based mode.

Maison Vella — E-Commerce Admin Panel — Next.js, tRPC, Prisma, Sanity

- Built a responsive e-commerce administration dashboard using Next.js 14, Server Actions, ShadCN UI, Tailwind CSS, and Clerk authentication.
- Implemented type-safe data access using **tRPC**, Prisma, Supabase/PostgreSQL, Sanity CMS, and Cloudflare R2 media storage.

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering

2022 – 2026

University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream)

2019

St. John's College, Panadura, Sri Lanka

Results: 3 A's (Combined Mathematics, Physics, Chemistry)

District Rank: 17

Training & Certifications

CKA Training — KodeKloud, 2025 • AWS SAA-C03 Training — Udemy, 2025 • Apache Kafka Series — Udemy, 2025 • 100 Days of DevOps — KodeKloud, 2025 • Advanced OAuth Security — Udemy, 2024

Articles

[See all articles](#)

- The Singleton Design Pattern in Java: From Beginner to Production-Ready** — Software Engineering / Java
- The Ultimate Guide to Software Optimization: Bentley's Rules Modernized** — Performance
- The Architecture of Trust: The Ultimate Guide to API Security** — API Security
- How Kafka Cheated the Two Generals' Problem to Achieve Exactly-Once Semantics** — Distributed Systems

Referees

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